

MAL 201 E - 12192

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Solid materials have been grouped in 3:

1-METALS

Composed of 1 or more metallic elements and often also relatively small amounts of non-metallic elements.

Stiff, strong, ductile, resistant to fracture

Denser than other materials

Have large amount of non-localized electrons. (conducts electric and heat)

2 - Ceramics

Compounds between metallic and nonmetallic elements
Oxide, nitride, carbides. (Al_2O_3 , SiO_2 , SiC , Si_3N_4)
Stiff, strong, brittle, susceptible to fracture, insulator

3 - Polymers

Many of them are organic compounds that are chemically based on C, H and other nonmetallic elements (O, N, Si)
Large molecular structures, often chain-like in nature that have a backbone of C atoms.

Low density, stiffness, strength
extremely ductile, pliable

Composites

Composed of two (or more) individual materials
Wood and bone are natural composites.

Advanced Materials

Semiconductors, biomaterials, smart materials

Atomic Bonding

Atomic mass unit = $\frac{1}{12}$ mass of $^{12}C = 1.66 \times 10^{-27}$ kg

Valence electrons determine Chemical - Electrical
Thermal - Optical properties.

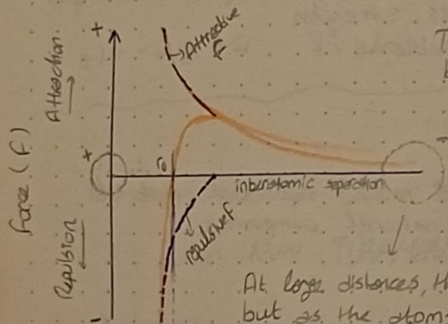
Primary Bonds (Ionic, Covalent, Metallic)

Secondary Bonds

A dipole may be created or induced in an atom or molecule that is normally electrically symmetric.

Permanent dipole moments exist in some molecules by virtue of an asymmetrical arrangement of positively and negatively charged regions.

Bonding Forces and Energies



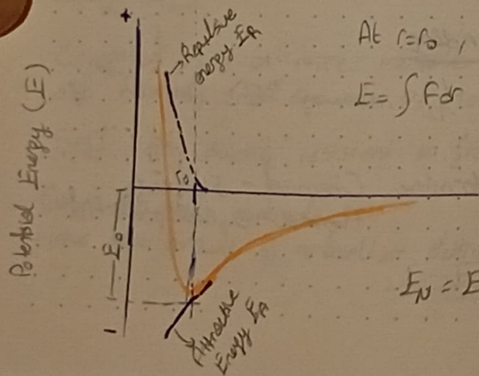
The principles of atomic bonding are best illustrated by considering the interaction between two isolated atoms as they are brought into close proximity from an infinite separation.

At large distances, the interactions are negligible, but as the atoms approach, each exerts forces on the other.

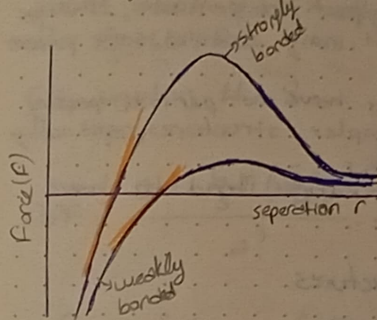
$$\text{At } r=r_0, F_A + F_R = 0$$

$$E = \int F dr = \int_{\infty}^{r_0} F_A dr + \int_{\infty}^{r_0} F_R dr$$

$$E_0 = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}$$



Materials having large bonding energies typically also have high melting temperatures.



The mechanical stress (modulus of elasticity) of a material depends on the curve.

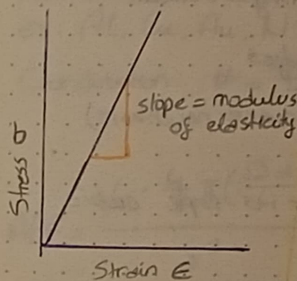
$$\frac{dF}{dr} \text{ at } r=r_0 \text{ will be}$$

steep for stiff, shallow for flexible materials

Melting Temperature is larger if E_0 is larger

$$\frac{\Delta L}{L_0} = \alpha (T_2 - T_1)$$

Larger if E_0 is smaller



Type	Energy	Comments
Ionic	Large	Nondirectional (ceramics)
Covalent	Variable large (diamond) small (bismuth)	Directional (semiconductors, ceramics, polymer chains)
Metallic	Variable large (Tungsten) small (mercury)	Nondirectional (metals)
Secondary	smallest	Directional inter-chain (polymer) inter-molecular

Ceramics (ionic, covalent)

Large bond energy.

Large T_m , E .

small α

Metals (metallic)

Variable bond energy.

Moderate T_m , E , α

Polymers (covalent, secondary) Directional properties, small T_m , E large α

The Structure of Crystalline Solids

Crystalline materials, atoms pack in periodic, 3D arrays typical of metals, many ceramics, some polymer

Noncrystalline materials, atoms have no periodic packing occurs for complex structures, rapid cooling

Dense, ordered, packed structures tend to have lower energies

Metallic Crystal Structures

- Simple Cubic Structure (SC)

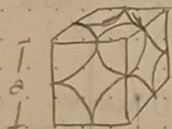
Rare due to low packing density. Polonium

Close-packed directions are cube edges

Coordination # = 6 (Num of nearest neighbors)

Atomic packing factor (APF) for a simple cubic structure = 0.52

$$APF = \frac{\text{Volume of atoms in unit cell}}{\text{Volume of unit cell}}$$



contains $8 \times \frac{1}{8} = 1$ atom/unit cell

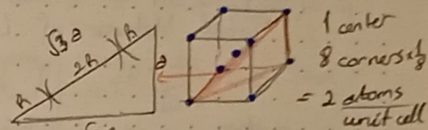
$$APF = \frac{1 \cdot \frac{4}{3} \pi (0.5a)^3}{a^3} = 0.52$$

- Body Centered Cubic Structure (BCC)

Atoms touch each other along cube diagonals.

Ex: Cr, W, Fe

Coordination # = 8



$$APF = \frac{2 \cdot \frac{4}{3} \pi \left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = 0.68$$

- Face Centered Cubic Structure (FCC)

Atoms touch each other along face diagonals.

ex: Al, Cu, Au, Ni, Ag

Coordination # = 12

$$6 \text{ face} \times \frac{1}{2} + 8 \text{ corners} \times \frac{1}{8}$$

= 4 atoms/unit cell

$$APF = \frac{4 \cdot \frac{4}{3} \pi \left(\frac{\sqrt{2}a}{4}\right)^3}{a^3} = 0.74$$

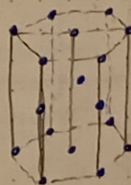
- Hexagonal Close-Packed Structure (HCP)

ex: Cd, Mg, Ti, Zn

Coordination # = 12

6 atoms/unit cell

$$APF = 0.74$$



Theoretical Density, ρ

$$\rho = \frac{\text{Mass of Atoms in Unit Cell}}{\text{Total Volume of Unit Cell}} = \frac{nA}{V_c N_A}$$

$$n = \frac{\# \text{ atoms}}{\text{unit cell}} \quad A = \text{atomic weight}$$

V_c = Volume of unit cell

Example

Cr (BCC)

$$A = 52.00 \text{ g/mol}$$

$$R = 0.125 \text{ nm}$$

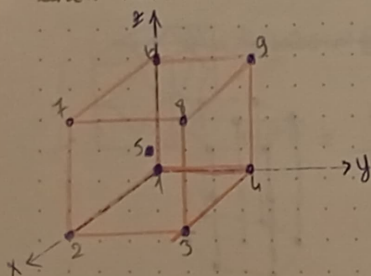
$$n = 2 \text{ atoms/unit cell}$$

$$a = 4R/\sqrt{3} = 0.2887 \text{ nm}$$

$$\rho = \frac{2 \cdot 52.00}{(0.2887)^3 \cdot 10^{-23}} = 7.18 \text{ g/cm}^3$$

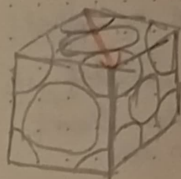
Crystal Systems

Point coordinates for all atom positions for a BCC unit cell

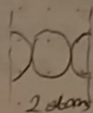


Point Number	Point Coordinate
1	000
2	100
3	110
4	010
5	$\frac{1}{2} \frac{1}{2} \frac{1}{2}$
6	001
7	101
8	111
9	011

$$\text{Linear Density of Atoms} = LD = \frac{\text{Number of Atoms}}{\text{Unit length of direction vector}}$$



linear density of Al in [110]



$$LD = \frac{2}{(2a)} = 3.5 \text{ nm}^{-1}$$

Example

Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminum and 3 wt% copper

$$C_{Al} = \frac{C_{Al} A_{Cu}}{C_{Al} A_{Cu} + C_{Cu} A_{Al}} \times 100 = \frac{97(63.55 \text{ g/mol})}{97(63.55 \text{ g/mol}) + 3(26.98 \text{ g/mol})} \times 100 = 98.7 \text{ at}\%$$

$$C_{Cu} = \frac{C_{Cu} A_{Cu}}{C_{Cu} A_{Cu} + C_{Al} A_{Al}} \times 100 = 1.30 \text{ at}\%$$

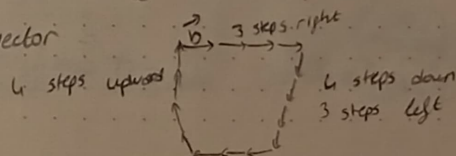
Imperfections in Solids

- Dislocations

• Edge Dislocation

Within the range around the dislocation line there is some localized lattice distortion

$\vec{b}$ = Burger's vector



Screw Dislocation

As being formed by a shear stress that is applied.



The atomic distortion associated with a screw dislocation is also linear and along a dislocation line.

Deformation Mechanisms

- Slip Plane

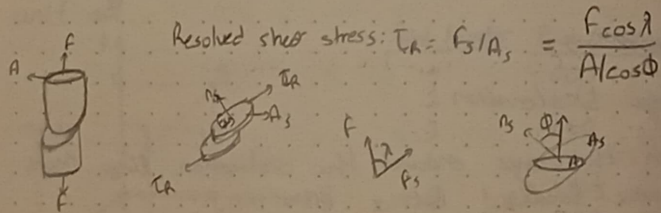
Plane on which easiest slippage occurs. Highest planar densities and large interplanar spacings.

- Slip Directions

Directions of movement. Highest linear densities.

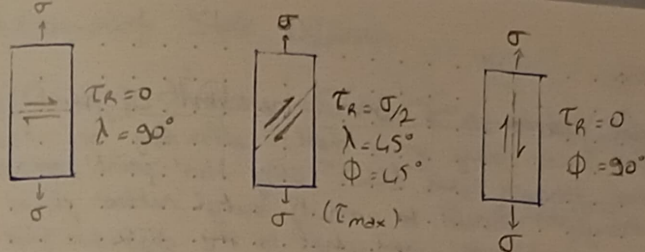
Stress and Dislocation Motion

- Resolved shear stress, τ_R , results from applied tensile stress



Applied tensile stress: $\sigma = F/A$

$$\tau_R = \sigma \cos \lambda \cos \phi$$



Condition for dislocation motion: $\tau_R > \tau_{CRSS}$

typically 10^4 GPa to 10^2 GPa

Diffusion Flux

$$J = \frac{\text{moles (or mass) diffusing}}{\text{surface area} \cdot \text{time}}$$

measured empirically

- Make thin film (membrane) of known surface area
- Impose concentration gradient
- Measure how fast atoms or molecules diffuse through the membrane

$$J = \frac{M}{At} = \frac{1}{A} \frac{dM}{dt}$$

→ Steady-State Diffusion

Rate of diffusion independent of time

Flux proportional to concentration gradient = $\frac{dC}{dx}$

$$J = -D \frac{dC}{dx}$$

diffusion coefficient $D = D_0 e^{\left(-\frac{Q_d}{RT}\right)}$

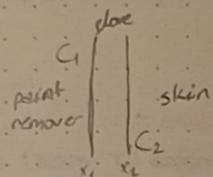
Example

Methylene chloride is a common ingredient of paint removers. Besides being an irritant, it also may be absorbed through skin. When using this paint remover, protective gloves should be worn. If butyl rubber gloves (0.04 cm thick) are used, what is the diffusive flux of "methylene chloride" through the glove?

$$D_{\text{butyl rubber}} = 110 \times 10^{-9} \text{ cm}^2/\text{s}$$

$$C_1 = 0.44 \text{ g/cm}^3$$

$$C_2 = 0.02 \text{ g/cm}^3$$



$$J = -D \frac{dC}{dx} \approx -D \frac{C_2 - C_1}{x_2 - x_1} = -(110 \times 10^{-9}) \frac{0.02 - 0.44}{0.04} \left[\frac{\text{g}}{\text{cm}^2 \cdot \text{s}} \right]$$

$$= 1.16 \times 10^{-5} \left[\frac{\text{g}}{\text{cm}^2 \cdot \text{s}} \right]$$

Example

At 300°C the diffusion coefficient and activation energy for Cu in Si are,

$$D(300^\circ\text{C}) = 7.8 \times 10^{-11} \text{ m}^2/\text{s}$$

$$Q_d = 41.5 \text{ kJ/mol}$$

What is the diffusion coefficient at 350°C?

$$\ln D_2 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T} \right) \quad \text{and} \quad \ln D_1 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_1} \right)$$

$$\ln \frac{D_2}{D_1} = -\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$T_1 = 573 \text{ K}$
 $T_2 = 623 \text{ K}$

$$D_2 = 15.7 \times 10^{-11} \text{ m}^2/\text{s}$$

→ Non-steady State Diffusion

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

$$\frac{C(x,t) - C_0}{C_s - C_0} = 1 - \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$

$\operatorname{erf}(z)$: error function

Example

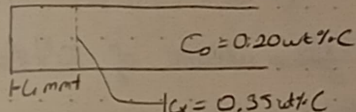
An FCC iron-carbon (Fe-C) alloy initially containing 0.20 wt% C is carburized at an elevated temperature and in an atmosphere that gives a surface carbon concentration constant at 1.0 wt%. It after 49.5 h the concentration of C is 0.35 wt% at a position 4.00 mm below the surface. Determine the temperature at which the treatment was carried out.

at $t=0$

$$C_s = 1.0 \text{ wt\% C} \quad C_0 = 0.20 \text{ wt\% C}$$

at $t = 49.5 \text{ h}$

$$C_s = 1.0 \text{ wt\% C}$$



$$\frac{C(x,t) - C_0}{C_s - C_0} = \frac{0.35 - 0.20}{1.0 - 0.20} = 1 - \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right) = 1 - \operatorname{erf}(z)$$

$$D = \frac{x^2}{4z^2 t} = \frac{(4 \times 10^{-3})^2 \text{ h}}{4(0.93)^2 (49.5) 3600 \text{ s}} = 2.6 \times 10^{-11} \frac{\text{m}^2}{\text{s}} \quad \operatorname{erf}(z) = 0.8125$$

$z = 0.93$

$$T = \frac{Q_d}{R(\ln D_0 - \ln D)} = 1300 \text{ K}$$

$\downarrow$
 2.3×10^{-5}

1- Specific gravity (density) and atomic radius of Au are given as 19.32 g/cm^3 and $1.44 \text{ \AA}$.

a. Calculate the # Au atoms in 1 mm^3 Au

Density, $\rho = 19.32 \text{ g/cm}^3 = 19.32 \times 10^{-3} \text{ g/mm}^3$

$1 \text{ mm}^3 \text{ Au} = 19.32 \times 10^{-3} \text{ g}$

Au = $\frac{19.32 \times 10^{-3}}{197 \text{ A.N.}} \approx 5.9 \times 10^{13} \frac{\text{atoms}}{\text{mm}^3}$

atomic weight

b. Calculate the # Au atoms in 1 g Au

$\frac{\text{A.N.}}{197} \approx 3 \times 10^{21} \frac{\text{atoms}}{\text{gram}}$

c. Calculate the lattice volume of 10^{21} Au atoms

$\frac{10^{21} \times 197 \text{ A.N.}}{19.32} = 0.01699 \text{ cm}^3 \approx 16.99 \text{ mm}^3$

d. Calculate the APF for Au

Volume of 1 Au atom = $\frac{4\pi R^3}{3}$ $R_{\text{Au}} = 1.44 \times 10^{-8} \text{ cm}$

$10^{21} \times V = 12.5 \text{ mm}^3$

APF = $\frac{\text{Volume of } 10^{21} \text{ atoms}}{\text{Lattice volume of } 10^{21} \text{ atoms}} = \frac{12.5}{16.99} \approx 0.74$

e. Decide if Au has FCC or BCC crystal structure

APF for FCC = 0.74 so FCC

2- Some gold (Au) is added in silver (Ag). The lattice constant and density of that FCC substitutional solid solution are 0.359 nm and 12.686 g/cm^3 . Determine the atomic percent of Au atoms.

For FCC: 4 atoms/unit cell

For this alloy: $\begin{cases} \# \text{ Au atoms/unit cell} = x \\ \# \text{ Ag atoms/unit cell} = 4 - x \end{cases}$
(substitutional solid solution)

$\rho = \frac{(4-x) \cdot 107.87 + x \cdot 197}{\text{A.N.} (0.359 \times 10^{-8})^3} = 12.686 \text{ g/cm}^3$ mass of 1 unit cell

$x = 0.98$ $\% \text{ Au} = \frac{0.98}{4} = 24.5\%$

$\% \text{ Ag} = \frac{4-0.98}{4} = 75.5\%$

3- (80 Ni - 20 Cu) substitutional solid solution has

FCC crystal structure. The atomic mass of

Ni and Cu are 58.71 and 63.54 g/mole

$R_{\text{Ni}} = 1.246 \text{ \AA}$ $R_{\text{Cu}} = 1.278 \text{ \AA}$

a. Calculate the average atomic radius in this alloy

For 100g alloy: 80g Ni + 20g Cu

$$\# \text{ Ni} = \frac{80}{(58.71/\text{A.N.})} = N_{\text{Ni}}$$

$$\# \text{ Cu} = \frac{20}{63.54/\text{A.N.}} = N_{\text{Cu}}$$

$$\% \text{ Ni} = \frac{N_{\text{Ni}}}{N_{\text{Total}}} = 81.2\%$$

$$\% \text{ Cu} = \frac{N_{\text{Cu}}}{N_{\text{Total}}} = 18.8\%$$

$$\text{Average atomic radius} = R_0 = [\% \text{ Ni} \times R_{\text{Ni}}] + [\% \text{ Cu} \times R_{\text{Cu}}]$$

$$R_0 = 0.812 \times 1.246 + 0.188 \times 1.278$$

$$R_0 = 1.252 \text{ \AA} = 1.252 \times 10^{-8} \text{ cm}$$

b. Calculate the average lattice constant

$$\text{For FCC} \Rightarrow a = \frac{4R_0}{\sqrt{2}} = \frac{4 \cdot 1.252 \times 10^{-8}}{\sqrt{2}}$$

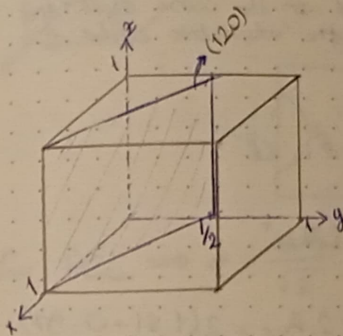
$$a = 3.54 \times 10^{-8} \text{ cm}$$

c. Calculate the density of this alloy

$$\text{Density} = \rho = \frac{4 \times (0.812 \times 58.71 + 0.188 \times 63.54)}{0.6 \times 10^{-24} \times (3.54 \times 10^{-8})^3}$$

$$= 8.96 \text{ g/cm}^3$$

d. Calculate the planar density of Cu atoms on (120) plane.

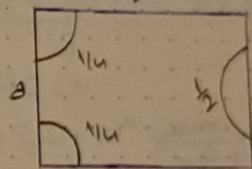


$$\rho_{(120)}^{\text{FCC}} = \frac{\# \text{ Atoms on (120)}}{\text{Plane area}}$$

$$= \frac{1}{\frac{a^2 \sqrt{5}}{2}} = \frac{2}{(3.54 \times 10^{-8})^2 \times \sqrt{5}} = 7.16 \times 10^{14} \text{ atoms/cm}^2$$

x	y	z
1	2	0
1/1	1/2	1/0
1	1/2	0

$$\frac{a\sqrt{3}}{2}$$



$$\# \text{ atoms} = (1)$$

$$\# \text{ Cu atoms on (120)} = \% \text{ Cu} \times \rho_{(120)} = 0.188 \times 7.16 \times 10^{14}$$

$$= 1.3 \times 10^{14} \text{ Cu} \left[\frac{\text{atoms}}{\text{cm}^2} \right]$$

4. Cubic unit cell of NaCl consists of 4 Na and 4 Cl atoms. Atomic mass of Na and Cl are 23 g/mole and 35.45 g/mole. Atomic radius of Na and Cl are $0.97 \text{ \AA}$ and $1.81 \text{ \AA}$. Determine theoretical density of NaCl.

In NaCl's complex cubic unit cell, Cl^- ions are located on the corners and face centers of the unit cell; Na^+ ions are located on the cube edges and one Na ion is located in the volumetric center of the cube.

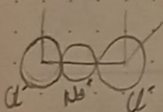
$$\# \text{Cl}^- = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4 \text{ Cl}^-$$

$$\# \text{Na}^+ = 12 \times \frac{1}{4} + 1 = 4 \text{ Na}^+$$

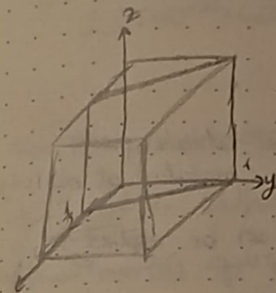
$$a = 2R_{\text{Na}} + 2R_{\text{Cl}} = 2(1.81 + 0.97) \\ = 5.56 \times 10^{-8} \text{ cm}$$

$$\rho = \frac{4(23.0 + 35.45)}{a^3 \times (5.56 \times 10^{-8})^3}$$

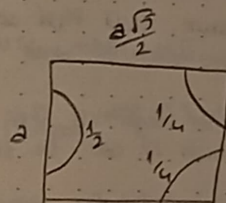
$$= 2.268 \text{ g/cm}^3$$



5. Calculate the planar density of (210) plane in Ag. Silver has FCC crystal structure and the atomic radius of silver is $1.444 \text{ \AA}$.



x	y	z
2	1	0
$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{2}$	1	0



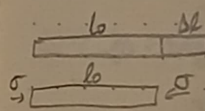
$$\# \text{ atoms} = \frac{1}{2} + 2 \times \frac{1}{4} = 1$$

$$\text{For FCC} \Rightarrow a = \frac{4R}{\sqrt{2}} = \frac{4 \times 1.444 \times 10^{-8}}{\sqrt{2}} = 4.085 \times 10^{-8} \text{ cm}$$

$$\rho_{(210)}^{\text{FCC}} = \frac{1}{\frac{(4.085 \times 10^{-8})^2}{2}} = 5.4 \times 10^{14} \frac{\text{atoms}}{\text{cm}^2}$$

Example

A rod is stress-free at 20°C . It's heated up, but prevented from lengthening. At what temperature the stress reach -172 MPa ?



Assume unconstrained thermal expansion

$$\frac{\Delta L}{L_{\text{room}}} = \epsilon_{\text{thermal}} = \alpha_L (T_f - T_0)$$

$$\text{Compress back to original} \quad \epsilon_{\text{compress}} = \frac{-\Delta L}{L_{\text{room}}} = -\epsilon_{\text{thermal}}$$

The thermal stress can be directly calculated as

$$\sigma = E(\epsilon_{\text{compress}}) = -E(\alpha_L(T_f - T_0))$$

$$T_f = T_0 - \frac{\sigma}{E\alpha_L} \rightarrow -172$$

$$20^\circ\text{C} \quad \downarrow \quad \begin{matrix} 100\text{GPa} \\ (brass rod) \end{matrix} \quad \downarrow \quad 20 \times 10^{-6}$$

$$T_f = 106^\circ\text{C}$$

Example

For a brass alloy the stress at which plastic deformation begins is 345 MPa and the modulus of elasticity is 103 GPa.

a) What is the maximum load that can be applied to a specimen with a cross-sectional area of 130 mm² without plastic deformation?

b) If the original specimen length is 76 mm, what is the maximum length to which it can be stretched without causing plastic deformation?

$$\sigma = \frac{P}{A} \quad \epsilon = \frac{\Delta L}{L} \quad E = \frac{\sigma}{\epsilon}$$

$$a) P = \sigma A = 345 \times 10^6 \times 130 \times 10^{-6} = 44,850 \text{ N}$$

$$103 \times 10^9 = \frac{345 \times 10^6}{\epsilon} \quad \epsilon = 3.35 \times 10^{-3} = \frac{\Delta L}{76 \times 10^{-3}}$$

Example

Strontium has an FCC crystal structure an atomic radius of 0.215 nm and an atomic weight of 87.62.

Calculate the theoretical density for Sr.

$$\rho = \frac{M}{V_c N_A}$$

FCC # atoms: 4

$$V_c = a^3 \rightarrow a\sqrt{2} = 4r \quad a = 2\sqrt{2}r \rightarrow 0.215 \text{ nm}$$

$$\rho = \frac{4 \times 87.62}{2.25 \times 10^{-22} \times 6.022 \times 10^{23}} = \frac{4 \times 87.62}{135.495} = 2.58 \text{ g/cm}^3$$

Example

The concentration of Si in a Fe-Si alloy is 0.25 wt %. What is the concentration in kilograms of Si per cubic meter of alloy?

$$\rho_{\text{iron}} = 7.87 \text{ g/cm}^3 \quad \rho_{\text{silicon}} = 2.33 \text{ g/cm}^3$$

1 g of alloy contains 0.25% Si = 0.0025 g

99.75% Fe = 0.9975 g

$$V_{\text{Si}} = \frac{0.0025}{2.33} = 1.0729 \times 10^{-3} \text{ cm}^3$$

$$V_{\text{Fe}} = \frac{0.9975}{7.87} = 0.1267 \text{ cm}^3$$

$$V_{\text{Si}} + V_{\text{Fe}} = 0.12777 \text{ cm}^3$$

$$C_{\text{Si}} = \frac{0.0025}{0.12777} = 0.0195678 \text{ g/cm}^3 = 19.567 \frac{\text{kg}}{\text{m}^3}$$

Example

A coil of wire 0.25m long and having 400 turns carries a current of 15 A.

(a) What is the magnitude of the magnetic field strength H ?

$$H = \frac{NI}{L} = \frac{400 \cdot 15}{0.25} = 24000 \text{ A/m}$$

(b) Compute the flux density B if the coil is in vacuum.

$$B = \mu_0 H = 1.257 \times 10^{-6} \cdot 24000 = 0.03 \text{ T}$$

(c) Compute the flux density inside a bar of chromium that is positioned within the coil. $\chi_m = 3.13 \times 10^{-4}$

$$B = \mu_0 H (1 + \chi_m) = 3.02 \times 10^{-2} \text{ T}$$

(d) Compute the magnitude of the magnetization M .

$$M = \chi_m H = 7.5 \text{ A/m}$$

Example

A steel wire is stretched with a stress of 70 MPa at 20°C. If the length is held constant, to what temperature must the wire be heated to reduce the stress to 17 MPa?

$$E = 207 \text{ GPa} \quad \alpha_L = 12 \times 10^{-6} / ^\circ\text{C}$$

$$\sigma = 70 - 17 = 53 \text{ MPa} \rightarrow \text{compressive, so } -53 \text{ MPa}$$

$$T_s = T_0 - \frac{\sigma}{E \alpha_L} = 20 - \frac{-53 \text{ MPa}}{\frac{207 \text{ GPa}}{10} \cdot 12 \times 10^{-6}} = 20 + 21.3 = 41.3^\circ\text{C}$$

